

# Cost–accuracy regimes for neural surrogate fidelity estimation of composed quantum channels

Shuchang Wang<sup>1,†</sup>, Xiaoman Liu<sup>1,†</sup> and Wei Yang<sup>1</sup>

<sup>1</sup> School of Computer Science and Technology, University of Science and Technology of China, Hefei 230026, China

E-mail: qubit@ustc.edu.cn

**Abstract.** Learned surrogates are often proposed as fast substitutes for scientific estimators, yet their advantage can disappear when a strong deterministic or measurement baseline is implemented. We study this issue for end-to-end entanglement fidelity of composed quantum channels. The benchmark spans hardware-like Markovian noise, a non-Markovian collision model, and two-qubit order-sensitive channels, and compares analytic approximations, exact superoperator composition, classical Monte Carlo, Direct Fidelity Estimation (DFE), low-capacity regressors, and neural sequence models under explicit information and label budgets. Two negative controls are decisive. When full per-step Choi matrices describe a Markovian sequence, direct composition is numerically exact (MAE about  $10^{-8}$ ) and takes 0.15–0.86 ms per sequence on CPU for the studied  $d \leq 4$  cases, so a learned replacement is not warranted. For the single-qubit collision target, enumerating the four Pauli settings and allocating shots across them removes the large protocol variance of sampling settings with replacement: stratified DFE reaches MAE 0.092 with 32 shots and 0.065 with 64 shots on the memory-strength shift. Neural models remain useful only in a narrower deployment regime. With 64 labelled out-of-distribution examples, a bidirectional sequence model reaches MAE 0.087 without per-query measurements, compared with 0.100 for a same-budget summary ridge model; when each calibration label is itself estimated using 64 DFE shots, performance is essentially unchanged and the 4096-shot offline cost is amortised after roughly 128 queries relative to 32-shot DFE. The result is a reproducible decision procedure: use exact composition when the complete Markovian description is available, use direct measurement when modest per-query shots are affordable, and consider a calibrated surrogate only for repeated measurement-constrained queries under a stable shift.

*Keywords:* surrogate models, out-of-distribution calibration, scientific machine learning, benchmarking, quantum channels, fidelity estimation, non-Markovian dynamics

<sup>†</sup> These authors contributed equally to this work.

## 1. Introduction

Learned surrogates are increasingly used as cheap stand-ins for expensive scientific computations, but it is often unclear when such a surrogate is actually preferable to the

cheaper analytic approximations, classical sampling estimators, or direct measurements it is meant to replace. The question matters most when the surrogate must generalise out of distribution and when its accuracy has to be weighed against a fixed measurement or simulation budget. We study this cost-accuracy tradeoff for one such setting: predicting the fidelity of composed quantum channels.

End-to-end fidelity prediction is a basic ingredient in several quantum-information workflows, including repeater design [1], purification-protocol selection [2], validation of deep NISQ circuits [3], and estimates of logical-qubit performance in quantum-error-correction studies [4]. In each case, the relevant quantity is the entanglement fidelity  $F_e$  of a composed channel obtained by chaining individual channels  $\Lambda_i$ . In practice, however, the available information is often marginal: one has descriptions of the individual channels, but not direct access to the full composed process. Exact composition also becomes costly as the Hilbert-space dimension and sequence length grow. This makes it natural to ask whether a learned surrogate can map a sequence of channel descriptions, such as Choi matrices, Pauli transfer matrices, or compact noise parameters, to the fidelity of the composed channel.

Several fidelity-estimation tools already exist, each with a different cost. The product approximation

$$\hat{F}_{\text{prod}} = \prod_{i=1}^n F_e(\Lambda_i) \quad (1)$$

requires no quantum measurements and is accurate when the composed dynamics are well described by sequential application of the marginal channels. We call it an approximation rather than a bound because its tightness depends on additional channel structure, such as Pauli diagonality; see Appendix Appendix A. Classical Monte-Carlo sampling over Kraus paths is also measurement-free, but can be computationally expensive. Direct Fidelity Estimation [5], henceforth DFE, instead measures the deployed composed channel and gives an unbiased estimate at a quantum-shot cost per sequence. Randomised benchmarking [6] provides another useful diagnostic, but estimates the average gate fidelity of Clifford operations rather than  $F_e$  for an arbitrary composed sequence. The key distinction is that product and marginal Monte-Carlo estimators only see per-channel information, whereas DFE probes the true composed process.

This distinction matters when consecutive channels are correlated through a shared environment. A neural surrogate can, in principle, learn such cross-channel corrections without requiring quantum measurements at inference time. Whether this is useful in practice depends on the operating regime. If the correction to the marginal prediction is smaller than the model-fitting error, learning has little to recover. If bath-mediated correlations dominate the fidelity, marginal estimators can fail, and a learned model may become useful. This paper studies that regime dependence directly, rather than treating fidelity prediction primarily as an architecture-selection problem.

We first derive an  $\varepsilon^2 L^2$  scaling diagnostic that relates the per-channel infidelity  $\varepsilon$  and sequence length  $L$  to the expected size of cross-channel corrections. The diagnostic

gives a direct criterion for choosing among analytic, classical Monte-Carlo, quantum-measurement, and learned estimators. It also identifies when a learned surrogate can improve on predictors that use only marginal channel information.

Guided by this diagnostic, we compare neural surrogates with analytic, Monte-Carlo, and DFE baselines in three regimes. In a hardware-calibrated device-noise regime whose single-qubit error rates are matched to `qiskit-aer FakeV2` device calibrations [7], the per-channel infidelity is around  $\varepsilon \approx 10^{-3}$ . Here the product approximation is already nearly exact, and the diagnostic correctly predicts that no learned model should be expected to improve on it: at these error rates and sequence lengths, the correlation correction is of the same order as the residual error of the analytic baseline itself. This is a decision the practitioner can make in advance, before training anything.

The situation changes for non-Markovian channel families. We construct a collision-model benchmark [8] with a bath-retention parameter  $\eta \in [0, 0.99]$ . At  $\eta = 0$ , the dynamics reduce to the Markovian marginal model and the product approximation is recovered. As  $\eta$  approaches one, the bath carries information across the sequence, and later collisions depend on earlier ones. In this correlation-dominated regime, marginal estimators saturate at high error because they are given only per-channel marginals and cannot access shared-bath correlations. A Choi-aware causal surrogate, followed by a lightweight affine recalibration on a small labelled out-of-distribution set, recovers much of the attainable accuracy while requiring no quantum measurements per deployment query. The labelled calibration set must itself be obtained from exact simulation, prior characterisation, or measurement on representative OOD instances; this is an offline cost, not a free zero-shot assumption. We compare the resulting amortised tradeoff explicitly against DFE in Section 5.4.

We also study a two-qubit setting with  $d = 4$ . There, learned surrogates can improve over the product approximation, but classical Monte-Carlo on marginal channels remains more accurate when that simulation budget is available. This regime therefore separates two deployment criteria: learned surrogates are attractive when both quantum measurements and classical Monte-Carlo sampling are too costly, but they are not a universal replacement for the best available estimator.

Our contributions are as follows:

- (a) We give an operating-regime characterisation of when learned surrogates help for composed-channel fidelity prediction, expressed through an  $\varepsilon^2 L^2$  scaling diagnostic.
- (b) We benchmark analytic estimators, classical Monte-Carlo, DFE, and neural surrogates across device-noise, non-Markovian, and two-qubit regimes, including the quantum-shot and calibration costs relevant to deployment.
- (c) We show that a simple affine recalibration on a labelled out-of-distribution set can recover a large fraction of the surrogate’s OOD accuracy in the non-Markovian regime.
- (d) We release a channel-composition benchmark with reproducible runs across the

three regimes, including a collision-model family designed to stress-test marginal-based fidelity estimators.

The main contribution is a regime map for deciding when learned fidelity predictors are useful, when analytic or Monte-Carlo estimators are preferable, and when quantum measurements remain necessary, rather than a particular neural architecture.

## 2. Setup

### 2.1. Channel sequences and entanglement fidelity

Let  $\mathcal{H}_S$  be a finite-dimensional system Hilbert space with dimension  $d$ . A quantum channel is a completely positive trace-preserving (CPTP) map  $\Lambda : \mathcal{B}(\mathcal{H}_S) \rightarrow \mathcal{B}(\mathcal{H}_S)$ . For an ordered sequence  $\Lambda_1, \dots, \Lambda_n$ , we write  $\Lambda_{1:n} \equiv \Lambda_n \circ \dots \circ \Lambda_1$ . We use the unnormalised Choi–Jamiołkowski matrix [9, 10]

$$\mathcal{J}(\Lambda) = \sum_{i,j=0}^{d-1} |i\rangle\langle j| \otimes \Lambda(|i\rangle\langle j|) \in \mathcal{B}(\mathcal{H}_S \otimes \mathcal{H}_S), \quad (2)$$

which obeys  $\mathcal{J}(\Lambda) \succeq 0$ ,  $\text{Tr}[\mathcal{J}(\Lambda)] = d$ , and  $\text{Tr}_2[\mathcal{J}(\Lambda)] = \mathbb{I}_S$  for every CPTP map.

The prediction target is the entanglement fidelity [11] of a channel with respect to the identity,

$$F_e(\Lambda) = \frac{1}{d^2} \text{Tr}[\mathcal{J}(\text{id})^\dagger \mathcal{J}(\Lambda)] = \langle \Phi^+ | (\text{id} \otimes \Lambda) (|\Phi^+\rangle\langle\Phi^+|) | \Phi^+ \rangle, \quad (3)$$

where  $|\Phi^+\rangle = d^{-1/2} \sum_i |ii\rangle$ . We use  $F_e$  rather than the average gate fidelity  $F_{\text{avg}} = (dF_e + 1)/(d + 1)$  because it is the target of our downstream estimates and of DFE.

The composed-channel prediction problem is

$$\text{FIDELITYFORMER}_\theta : (\mathcal{J}(\Lambda_1), \dots, \mathcal{J}(\Lambda_n)) \mapsto p_\theta(F_e(\Lambda_{1:n})), \quad (4)$$

where  $p_\theta$  is a calibrated distribution over  $F_e \in [0, 1]$ , implemented with a quantile head.

### 2.2. Three operating regimes

We evaluate the estimators in three regimes spanning more than two orders of magnitude in per-channel infidelity.

**(R1) Device regime** ( $d=2$ ,  $\varepsilon \approx 10^{-3}$ ). Single-qubit channels are sampled from hardware-calibrated parametric noise models spanning amplitude damping, phase damping, depolarising, and Lindblad families, with parameter ranges matched to the reported single-qubit gate-error scale of current superconducting and trapped-ion processors [7]. The `family_ood` split holds out the Pauli-channel family entirely. The per-channel infidelity is  $\varepsilon \approx 10^{-3}$ , matching the hardware gate-error scale.

**(R2) Non-Markovian collision regime** ( $d=2$ ,  $\varepsilon \approx 0.4$ ). The system and a bath qubit initially in  $|+\rangle$  interact through step-dependent commuting  $ZZ$  Hamiltonians. After each collision the joint state is retained with probability  $\eta$  and otherwise the bath is reset. Training, calibration, and ID data use  $\eta \in [0, 0.7]$ ; the legacy `family_ood` file

instead holds out the stronger interval  $\eta \in [0.85, 0.99]$ . It is therefore a memory-strength distribution shift, not a held-out collision family.

**(R3) Two-qubit order-sensitive regime ( $d=4$ ).** This regime contains two-qubit channels built from correlated dephasing, two-qubit depolarising noise, and imperfect SWAP/CNOT operations with both coherent and incoherent errors. The order-sensitive variant introduces non-commutativity between consecutive channels, so order-invariant baselines such as DeepSets lose relevant information. This regime tests whether the behaviour observed at  $d = 2$  extends to larger systems.

### 2.3. Splits and metrics

Each regime is divided into `train`, `calib`, `id_test`, `length_ood`, and a distribution-shift split. The first three use lengths in  $\{2, 4, 8, 16\}$  and `length_ood` uses  $\{24, 32, 48\}$ . R1 and R3 hold out a noise family; R2 holds out a high- $\eta$  memory-strength interval while retaining the legacy filename `family_ood` for code compatibility.

The main metric is pooled MAE on  $F_e$  across five independent seed-controlled splits. We also report the MAE after post-hoc affine recalibration (Appendix Appendix B) and the quantum-shot cost for measurement-based estimators. Calibration metrics for the distributional outputs, including CRPS, expected calibration error (ECE), and quantile coverage, are reported in the supplementary material. Unless stated otherwise, tables report means over five seeds, with standard deviations shown alongside the neural rows; per-seed recalibration coefficients are given in the appendix.

## 3. Architecture

### 3.1. Choi-aware per-channel encoder

Each channel  $\Lambda_i$  enters the model through its Choi matrix  $\mathcal{J}(\Lambda_i) \in \mathbb{C}^{d^2 \times d^2}$ . The Choi representation preserves CPTP structure (positivity and the partial-trace constraint) exactly, unlike a flattened superoperator. We Hermitian-symmetrise  $\mathcal{J}(\Lambda_i) \rightarrow \frac{1}{2}(\mathcal{J}(\Lambda_i) + \mathcal{J}(\Lambda_i)^\dagger)$  and split the result into real and imaginary parts, yielding  $2d^4$  real numbers per channel (32 for  $d=2$ , 512 for  $d=4$ ). The imaginary block is anti-symmetric and therefore formally over-parameterised; we retain it for code simplicity and let the encoder learn the redundancy. A two-layer MLP (multi-layer perceptron) with width  $D = 256$  and GELU activations maps the vector to a channel token  $z_i \in \mathbb{R}^D$ . As an ablation we provide a Pauli-transfer-matrix (PTM) variant of the encoder, which produces  $d^4 = 16$  real numbers per channel at  $d=2$ . Supplementary Sec. S4 finds PTM and Choi encoders statistically indistinguishable on the collision regime, with differences below 0.003 pooled MAE on all three splits and within seed noise.

### 3.2. Causal transformer backbone

A depth-6 causal transformer with  $h = 8$  heads,  $d_{\text{model}} = D = 256$ , FFN expansion  $4\times$ , and rotary positional encoding then processes the channel tokens  $z_1, \dots, z_n$ . Causal masking blocks the position- $i$  token from attending to later channels, which we find beneficial for stable length-OOD behaviour: bidirectional attention overfits the within-distribution length distribution and tends to degrade faster on extrapolation, even though it slightly outperforms causal attention on `id_test` and the two variants are close on the calibrated OOD splits (Section 5). We adopt causal masking because it matches the sequential composition structure and is the more stable default, not because it is uniquely required.

### 3.3. Quantile distribution head

The pooled sequence representation feeds two heads. The primary head produces  $K = 9$  quantile predictions at nominal levels  $\{0.1, 0.2, \dots, 0.9\}$ , trained with the pinball (quantile-regression) loss

$$\mathcal{L}_{\text{pinball}} = \sum_{k=1}^K \rho_{q_k}(F_e^{\text{true}} - \hat{F}_{q_k}), \quad (5)$$

with  $\rho_q(u) = u(q - \mathbb{K}[u < 0])$ . We initialise the median-quantile head bias so that the softplus output starts at  $F_e \approx 0.7$ ; without this initialisation the head saturates into a dead-gradient region within the first few hundred steps when fitting the deep-circuit regime. As an ablation, we also implement a Gaussian head outputting  $(\mu, \log \sigma)$  trained with truncated-normal NLL.

### 3.4. Auxiliary physics-consistency loss

We attach a small auxiliary head (weight  $\lambda_{\text{aux}} = 0.1$ ) that predicts the purity  $\text{Tr}[\mathcal{J}^2]/d^2$  of the composed Choi, intended to encourage internal representations that separate unitary from heavily decoherent compositions. The auxiliary head does not affect point-prediction MAE on the collision regime ( $\lambda_{\text{aux}} \in \{0, 0.1\}$  match within seed noise on all three splits); consistent with the theme of this paper, no single architectural component is decisive, and the Choi-aware encoder and causal backbone are best understood as reasonable, stable defaults rather than as the source of the effect. Full design and the ablation are in the supplementary material; the head is removed at inference.

### 3.5. Post-hoc affine recalibration

Distribution shift between training and OOD splits manifests as a near-affine bias in the predicted median:  $\hat{F}_{q_{0.5}}^{\text{raw}} \approx a^{-1}(F_e^{\text{true}} - b)$  over the OOD population. We exploit this with a one-line correction

$$\hat{F}_{q_{0.5}}^{\text{cal}} = a \cdot \hat{F}_{q_{0.5}}^{\text{raw}} + b, \quad (6)$$

where  $(a, b)$  are fitted by ordinary least squares on a labelled OOD calibration set disjoint from the test set. Our headline configuration uses  $N_{\text{cal}} = 64$  because the calibration-size sweep saturates at  $N_{\text{cal}} \in \{64, 128\}$  (Fig. 2; a sweep down to  $N_{\text{cal}} = 8$  appears in Section 5.2). We call this the *labelled-ODD calibration* setting: the test set sees no labels at inference, but a labelled OOD calibration set is assumed available offline. Synthetic labels are generated by the corresponding exact simulator: Choi/superoperator composition for the Markovian regimes and joint system-bath propagation for the collision regime. In an experimental deployment they would have to come from an offline labelled characterisation campaign, such as DFE or tomography on representative OOD instances, and their quantum or classical cost must be amortised over subsequent deployment queries. The result is zero *per-query* measurement cost after calibration, not unconditional zero-shot learning.

We use a linear rather than isotonic, conformal, or non-linear correction because the residual after subtracting  $(a, b)$  from the raw predictions is small relative to the OOD bias (Appendix Appendix B). The same correction is applied at every quantile level for the distributional output.

Across five seeds, the raw  $(a, b)$  vary by  $\text{std} \approx 0.09$ , but the post-calibration MAE varies by only  $1.3 \times 10^{-3}$ , a factor of  $\approx 70$  smaller than the bias correction itself (Appendix Appendix B). The seeds reach the same calibrated MAE along several distinct affine paths, indicating that the OOD bias is structural rather than seed noise.

## 4. Baselines

We compare FIDELITYFORMER with analytic estimators, measurement-based estimators, classical Monte-Carlo sampling, and neural alternatives.

### 4.1. Analytic baselines

*Product approximation.* The primary analytic baseline is  $\hat{F}_{\text{prod}} = \prod_i F_e(\Lambda_i)$ , computed directly from the per-channel Choi matrices as in (1). It uses only marginal channel information and requires no quantum measurements.

*Exact composition of observed channels.* When full per-step Choi matrices are supplied and the process is Markovian, their superoperators can be multiplied deterministically and the target fidelity evaluated exactly. We include this baseline and its wall-clock cost. In R2 it composes only the reset-bath marginals, so its residual error measures the information missing from those inputs.

*Fuchs-van de Graaf bound.* We also use a conservative estimator derived from the relation between fidelity and Choi trace distance,

$$F_e \geq 1 - \frac{1}{2} \|\text{id} - \Lambda\|_{1,1}.$$

Per-channel infidelities are propagated through a sub-additive composition rule to obtain a one-sided lower bound on the composed-channel fidelity. We report its MAE as a point predictor to quantify how well this cheap analytic lower bound tracks the true  $F_e$ .

*Diamond-norm SDP.* As a stronger analytic baseline, we solve the Watrous diamond-norm SDP [12] for the marginal-composed channel and convert the result using

$$F_e \geq 1 - \frac{1}{2} \|\text{id} - \Lambda\|_\diamond.$$

The SDP is solved with SCS through `cvxpy` [13] and the QuTiP wrapper [14]. Like the product approximation, this baseline uses the marginal-composed channel, not the true non-Markovian composed process.

#### 4.2. Measurement and sampling baselines

*Direct Fidelity Estimation [5].* DFE estimates  $F_e(\Lambda_{1:n})$  by measuring Pauli observables on the deployed composed channel. For the single-qubit identity target all four Pauli settings have nonzero relevance. Sampling those settings with replacement adds avoidable protocol variance once the requested number of settings exceeds four. We therefore enumerate the support, allocate a fixed total shot budget across it in proportion to the square root of the target relevance, and simulate exact binomial  $\{+1, -1\}$  outcomes. We report total projective shots per sequence, sweeping 4–2000 shots in R2.

*Monte-Carlo Kraus sampling.* This classical baseline samples  $K$  Kraus paths from the per-channel marginal channels, propagates the maximally entangled state along each path, and averages the resulting fidelities [15]. We sweep  $K \in \{10, 100, 1000\}$ . Since the sampling distribution is built from marginal channels, this estimator is blind to correlations carried by a shared bath.

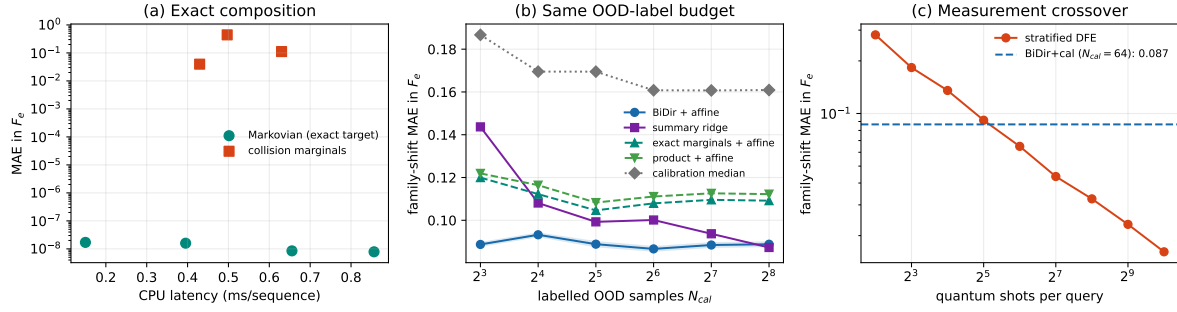
*Randomized benchmarking.* Randomized benchmarking [6] is not included in the direct cost-accuracy comparison because it estimates average gate fidelity from twirled Clifford sequences rather than the entanglement fidelity of a fixed arbitrary composed channel. Its cost is also tied to decay-curve estimation rather than to a per-sequence query model. We discuss its relationship to our setting in Section 7.

#### 4.3. Neural baselines

We train four neural alternatives using the same data, channel encoder, distribution head, optimiser, schedule, and seed list as FIDELITYFORMER.

**MLP.** A positional MLP that concatenates the Choi features from all channels and is therefore order-aware.





**Figure 1. Negative controls and corrected cost-accuracy comparison.** (a) Direct composition is numerically exact on Markovian full-Choi inputs at the studied  $d \leq 4$  scale; collision points show the error of composing only observable reset-bath marginals. (b) All OOD-calibrated methods receive identical labelled indices. (c) Stratified DFE reaches the calibrated neural error near 32 shots per query and surpasses it at 64 shots.

**DeepSets** [16]. An order-invariant model that sum-pools channel tokens. We include it to test whether the regime contains information that depends on channel order.

**BiDir Transformer.** A transformer with the same backbone as FIDELITYFORMER but without the causal mask. This isolates the effect of causal attention.

**GNN.** A linear-chain graph neural network with channel tokens as nodes and edges between consecutive channels. We use two message-passing variants, GNN and GENERIC\_GNN, both built on the shared channel encoder.

All neural models are trained with AdamW, learning rate  $3 \times 10^{-4}$ , cosine decay, 80 epochs, batch size 256, and early-stopping patience 20. The neural baselines differ only in sequence backbone and order handling.

## 5. Experiments

We present the results by operating regime. Figure 1 gives the cost-accuracy summary, and Tables 1 to 3 report the corresponding MAE values.

### 5.1. Device regime: hardware-realistic noise

Table 1 reports the device-noise regime. Because the input contains the full Choi matrix of every Markovian step, direct superoperator composition is the decisive control: its MAE is  $2.4 \times 10^{-8}$ ,  $4.2 \times 10^{-8}$ , and  $3.8 \times 10^{-8}$  on ID, length-OOD, and family-OOD, respectively, consistent with float32 feature roundoff. No learned estimator is justified when this information and computation are available. The product approximation remains the cheapest closed-form alternative and is accurate at  $10^{-4}$ – $10^{-3}$  scale. The near-zero marginal diamond-SDP value on family-OOD is a Pauli-geometry artefact, not a generic SDP advantage.

For nearly Pauli-diagonal channels, the correction missed by the product approximation scales as  $\varepsilon^2 L^2$ . At  $\varepsilon \approx 10^{-3}$  and  $L = 16$ , this gives a correction of

**Table 1. Device regime: pooled MAE on  $F_e$ .** Exact composition uses 1024 sequences per split; neural rows use five seeds and labelled-OOD calibration. The full-Choi exact baseline is limited only by stored float32 precision.

| Method                                                                          | id_test                          | length_ood                     | family_ood                     |
|---------------------------------------------------------------------------------|----------------------------------|--------------------------------|--------------------------------|
| <i>Deterministic and analytic, zero shots</i>                                   |                                  |                                |                                |
| Exact composition                                                               | $2.4 \times 10^{-8}$             | $4.2 \times 10^{-8}$           | $3.8 \times 10^{-8}$           |
| Product approx.                                                                 | $1.0 \times 10^{-4}$             | $1.6 \times 10^{-3}$           | $2.8 \times 10^{-4}$           |
| FvG bound                                                                       | $3.5 \times 10^{-4}$             | $5.4 \times 10^{-3}$           | $1.2 \times 10^{-3}$           |
| Diamond SDP                                                                     | $1.0 \times 10^{-2}$             | $5.1 \times 10^{-2}$           | $7.8 \times 10^{-8}^\dagger$   |
| <i>Neural, no per-query quantum measurements after labelled-OOD calibration</i> |                                  |                                |                                |
| FIDELITYFORMER+ cal                                                             | $(4 \pm 2) \times 10^{-4}$       | $(4.3 \pm 0.1) \times 10^{-2}$ | $(1.1 \pm 0.2) \times 10^{-2}$ |
| MLP + cal                                                                       | $(6.8 \pm 0.9) \times 10^{-3}$   | $(5.0 \pm 1.4) \times 10^{-2}$ | $(1.4 \pm 0.6) \times 10^{-2}$ |
| DEEPSETS + cal                                                                  | $(1.19 \pm 0.00) \times 10^{-2}$ | $(6.4 \pm 0.0) \times 10^{-2}$ | $(2.2 \pm 0.0) \times 10^{-2}$ |
| BIDIR + cal                                                                     | $(7 \pm 2) \times 10^{-4}$       | $(3.5 \pm 0.2) \times 10^{-2}$ | $(1.5 \pm 0.3) \times 10^{-2}$ |
| <i>Measurement and sampling baselines</i>                                       |                                  |                                |                                |
| MC, $K=10$ (0 q-shots)                                                          | $2.6 \times 10^{-2}$             | $6.9 \times 10^{-2}$           | $4.8 \times 10^{-2}$           |
| MC, $K=100$ (0 q-shots)                                                         | $9.7 \times 10^{-3}$             | $2.1 \times 10^{-2}$           | $1.5 \times 10^{-2}$           |
| MC, $K=1000$ (0 q-shots)                                                        | $3.1 \times 10^{-3}$             | $6.7 \times 10^{-3}$           | $4.8 \times 10^{-3}$           |

<sup>†</sup>In this split the held-out families leave a Pauli-channel test case. For Pauli channels, the relevant Fuchs-van de Graaf/diamond-norm bound is saturated up to solver tolerance, and Pauli structure is preserved under composition. This near-zero value is therefore a property of the split geometry, not a generic advantage of the SDP estimator.

order  $2.5 \times 10^{-4}$ , comparable to the residual error of the product approximation and well below the finite-sample fitting error of the neural models. The diagnostic therefore identifies this regime, in advance, as one where the analytic estimator is the appropriate choice and additional model complexity is not warranted.

## 5.2. Non-Markovian collision regime

The collision experiment is an information-limited stress test. The model receives reset-bath marginal Choi matrices, whereas the label is computed by propagating the joint system-bath state. The retention coefficient  $\eta$  is sampled independently and is not included in the input; for fixed collision parameters, changing  $\eta$  leaves every input marginal exactly unchanged. Consequently no estimator using these inputs can identify the realised  $\eta$  of an individual sample. It can only learn how the memory correction varies with the observable collision sequence, averaged over the training distribution, and a labelled deployment set can update that conditional average after a shift. We therefore call the legacy file `family_ood` a *memory-strength OOD* split: training uses  $\eta \in [0, 0.7]$  and the held-out split uses  $\eta \in [0.85, 0.99]$ .

Table 2 separates three comparisons that were conflated by weaker baselines. First, exact composition of the observed marginals is a deterministic zero-shot baseline; its memory-strength-OOD MAE is 0.442, confirming that the missing variable is bath

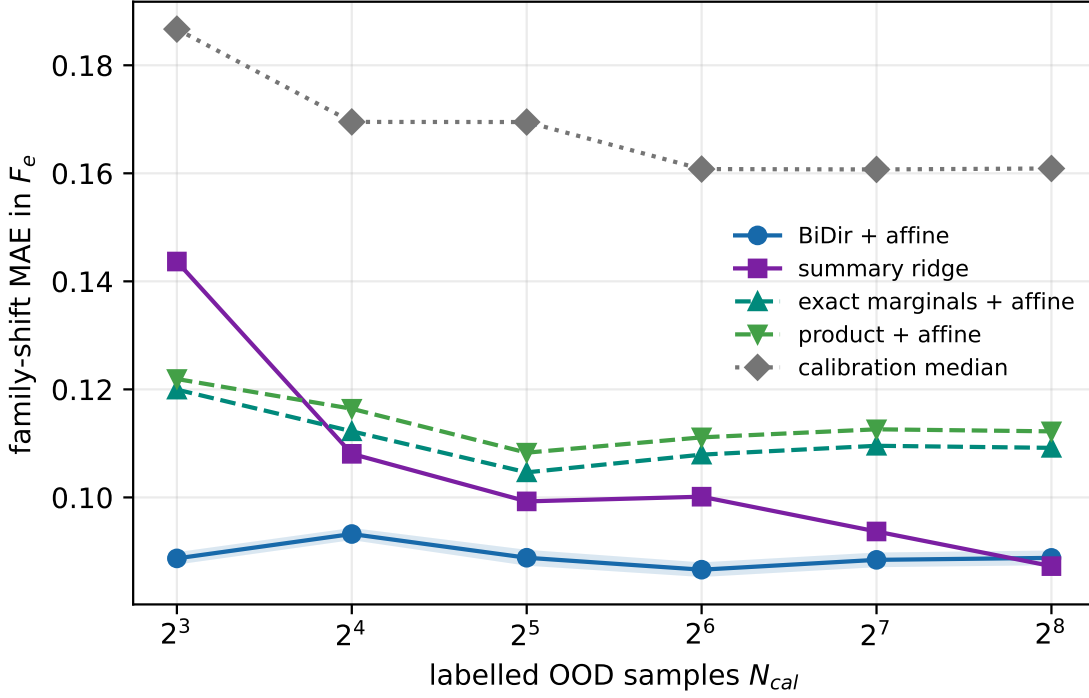
memory rather than numerical composition error. Second, every label-assisted method receives the same OOD calibration indices. At  $N_{\text{cal}} = 64$ , affine-calibrated product and exact-marginal predictors reach MAE 0.111 and 0.108, an eight-feature ridge model reaches 0.100, and the bidirectional neural model reaches 0.087. The neural gain is therefore real at small label budget, but smaller than a comparison against the uncalibrated product alone suggests. At  $N_{\text{cal}} = 256$ , the summary ridge reaches 0.087 and is statistically tied with the neural model.

Third, the measurement baseline must exploit the small Pauli support of this target. For a single-qubit identity target there are only four Pauli settings. Our revised DFE enumerates those settings, allocates a fixed total shot budget across them, and samples exact binomial outcomes. It reaches MAE 0.092 with 32 shots per sequence and 0.065 with 64 shots, so DFE is the accuracy-preferred method whenever even a modest per-query measurement budget is available. The calibrated surrogate instead targets repeated queries for which per-query measurements are disallowed. Using 64 calibration labels estimated at 64 shots each costs 4096 shots offline and yields MAE 0.086–0.089 across seeds, essentially the same as exact-label calibration. Relative to 32-shot DFE, this cost is amortised after about 128 deployment queries, although 64-shot DFE remains more accurate.

**Table 2. Collision regime: MAE on  $F_e$ .** Neural and low-capacity rows use the same  $N_{\text{cal}} = 64$  labelled memory-strength-OOD indices. DFE uses stratified allocation with exact binomial shot simulation. Dashes denote settings not rerun in the low-shot audit.

| Method                             | id_test | length_ood | memory-strength OOD                 |
|------------------------------------|---------|------------|-------------------------------------|
| Product approximation (0 shots)    | 0.038   | 0.086      | 0.431                               |
| Exact observed marginals (0 shots) | 0.040   | 0.112      | 0.442                               |
| MC, $K=1000$ (0 q-shots)           | 0.041   | 0.112      | 0.437                               |
| Calibration median                 | –       | –          | 0.161                               |
| Product + affine                   | –       | –          | 0.111                               |
| Exact marginals + affine           | –       | –          | 0.108                               |
| Summary ridge                      | –       | –          | 0.100                               |
| FIDELITYFORMER + affine            | 0.025   | 0.077      | $0.091 \pm 0.001$                   |
| BIDIR + affine                     | 0.025   | 0.072      | <b><math>0.087 \pm 0.001</math></b> |
| Stratified DFE, 32 shots           | –       | –          | 0.092                               |
| Stratified DFE, 64 shots           | –       | –          | 0.065                               |
| Stratified DFE, 2000 shots         | 0.0069  | 0.0112     | 0.0116                              |

*Architecture comparison.* The bidirectional transformer is slightly more accurate than the causal model after calibration, while the difference is small relative to the gap between either sequence model and the label-only controls. We do not interpret the weaker DeepSets result as evidence of order dependence in the present collision



**Figure 2. Same-budget OOD calibration sweep.** All curves use identical labelled memory-strength-OOD indices. The neural advantage is largest at small  $N_{cal}$ ; the eight-feature ridge baseline catches up near  $N_{cal} = 256$ .

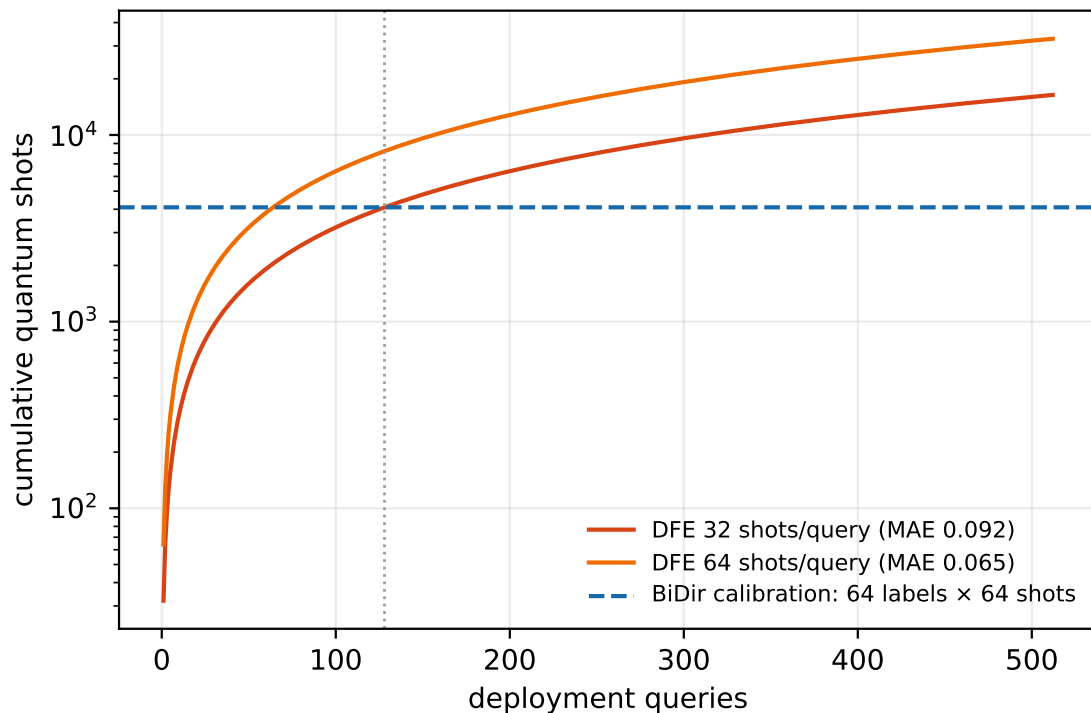
generator: its commuting  $ZZ$  interactions do not provide a clean order intervention. Order sensitivity is tested separately in the non-commuting  $d = 4$  benchmark.

### 5.3. Two-qubit $d=4$ scaling

Table 3 reports results for the two-qubit order-sensitive regime. Here FIDELITY-FORMER+cal improves over the product approximation by factors of 1.9, 2.3, and 1.8 on `id_test`, `length_ood`, and `family_ood`, respectively. The order-aware neural models also outperform the closed-form analytic baselines across the three splits.

This advantage does not extend to classical Monte-Carlo when a marginal simulation budget is available. MC with  $K = 100$  already reaches MAE 0.021 on `id_test`, and  $K = 1000$  reduces this to 0.007, well below the neural models. Thus the  $d = 4$  experiment supports the regime interpretation rather than a general learned-surrogate advantage: neural predictors are appropriate when analytic formulas are too rough and per-query simulation or measurement is too costly, but Monte-Carlo remains preferable when marginal simulation is cheap enough.

*Higher-dimensional regime pattern.* The  $d = 4$  results show that closed-form analytic baselines lose accuracy as the channel family becomes richer and more order-sensitive. They also show the limit of the learned approach: when classical marginal Monte-Carlo



**Figure 3. Amortised measurement cost.** Acquiring 64 calibration labels at 64 shots each costs 4096 shots once. This equals 128 queries of 32-shot DFE (MAE 0.092, compared with about 0.087 for the calibrated model). A 64-shot DFE query is more accurate (MAE 0.065) and crosses the same offline cost after 64 queries.

**Table 3.  $d=4$  two-qubit order-sensitive: pooled MAE on  $F_e$ .**

| Method               | id_test                             | length_ood                          | family_ood                          |
|----------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Product approx.      | 0.110                               | 0.076                               | 0.091                               |
| FvG bound            | 0.238                               | 0.102                               | 0.218                               |
| FIDELITYFORMER + cal | <b>0.059 <math>\pm</math> 0.002</b> | 0.033 $\pm$ 0.000                   | 0.051 $\pm$ 0.001                   |
| MLP + cal            | 0.075 $\pm$ 0.001                   | 0.038 $\pm$ 0.001                   | 0.060 $\pm$ 0.001                   |
| DEEPSSETS + cal      | 0.135 $\pm$ 0.002                   | 0.038 $\pm$ 0.001                   | 0.107 $\pm$ 0.002                   |
| BIDIR + cal          | 0.074 $\pm$ 0.000                   | <b>0.032 <math>\pm</math> 0.001</b> | 0.055 $\pm$ 0.002                   |
| GNN + cal            | 0.077 $\pm$ 0.003                   | 0.033 $\pm$ 0.001                   | 0.047 $\pm$ 0.001                   |
| GENERIC_GNN + cal    | 0.072 $\pm$ 0.003                   | 0.033 $\pm$ 0.001                   | <b>0.039 <math>\pm</math> 0.002</b> |
| MC, $K=10$           | 0.069                               | 0.056                               | 0.070                               |
| MC, $K=100$          | 0.021                               | 0.021                               | 0.022                               |
| MC, $K=1000$         | 0.007                               | 0.007                               | 0.007                               |

is affordable, it is simpler and more accurate.

*GNN under family shift.* The `GENERIC_GNN` model gives the best `family_ood` MAE in this regime (0.039 versus 0.051 for `FIDELITYFORMER`). We treat this as evidence that a path-graph inductive bias can help under some distribution shifts, but do not draw a broader architectural conclusion from a single split.

#### 5.4. Sample-complexity crossover

Figure 1(c) gives the corrected measurement crossover. Stratified DFE reaches the calibrated neural error at approximately 32 shots per query, not thousands of shots. Increasing the budget to 64 shots reduces MAE to 0.065, below every zero-measurement learned or marginal estimator in the memory-strength-OOD split. The surrogate is therefore not an accuracy replacement for DFE; it is an amortisation option when the deployment contract forbids measurements on each query. If 64 finite-shot calibration labels are collected with 64 shots each, the 4096-shot build cost is recovered after roughly 128 queries relative to 32-shot DFE. This calculation excludes the synthetic training-set generation cost, which is classical in this benchmark but could dominate if training labels also required hardware measurements.

#### 5.5. Wall-clock latency

Table 4 adds the deterministic baseline omitted from the original comparison. On the same server, direct superoperator composition of full Choi inputs is faster than the neural model for the single-qubit ID setting and of the same order for  $d = 4$ . Exact composition remains numerically exact for the Markovian datasets. The timings use CPU for exact composition and GPU for neural inference and should not be read as a hardware-normalised speed contest; their purpose is to rule out a surrogate-speed claim for the studied small dimensions.

**Table 4. Per-sequence wall-clock time (ms),  $n \leq 16$ .** Exact composition uses CPU; neural inference uses one RTX 5090.

| Method                                       | ms/sequence         |
|----------------------------------------------|---------------------|
| Product approximation                        | $0.0013 \pm 0.0003$ |
| Exact composition, $d = 2$                   | 0.150               |
| <code>FIDELITYFORMER</code> (small), $d = 2$ | $0.43 \pm 0.02$     |
| Exact composition, $d = 4$                   | 0.655               |
| MC, $K=10$                                   | $3.45 \pm 0.10$     |
| MC, $K=100$                                  | $30.2 \pm 0.36$     |
| MC, $K=1000$                                 | $298.9 \pm 4.6$     |

### 5.6. Distribution calibration: ECE and CRPS

Table 5 reports distribution-quality metrics for the collision regime before the labelled-ODD affine correction used in Table 2. The MAE column is therefore the pre-affine point-prediction MAE, matching the raw FIDELITYFORMER row up to rounding, while ECE and CRPS evaluate the conformally adjusted quantile output. On the in-distribution split the quantile output achieves ECE 0.0042 and CRPS 0.018, close to the ideal calibration target. On length-ODD and family-ODD, however, the ID-fitted quantile correction does not transfer: ECE rises to 0.39 and 0.50 respectively, while CRPS deteriorates in line with the raw MAE. The separate labelled-ODD affine recalibration in Table 2 corrects the mean MAE but does not recalibrate the full predictive distribution; we leave isotonic or conformal recalibration of the full OOD distribution to future work.

**Table 5. Collision regime: pre-affine distribution metrics for FidelityFormer (pooled over 5 seeds, 1024 sequences).** MAE is the raw/pre-affine point error; ECE is measured on the predicted median band and CRPS over the nine-quantile output after ID-fitted quantile correction.

| Split      | MAE   | ECE          | CRPS  |
|------------|-------|--------------|-------|
| id_test    | 0.025 | <b>0.004</b> | 0.018 |
| length_ood | 0.138 | 0.394        | 0.119 |
| family_ood | 0.383 | 0.500        | 0.359 |

## 6. Operating Regimes and Implications

The empirical pattern in Section 5 yields a practical estimator-selection rule. The rule is based on a scaling argument and on the regimes studied here, so its numerical thresholds should be treated as calibration points rather than universal constants.

### 6.1. The $\varepsilon^2 L^2$ scaling argument

For a Pauli-diagonal channel  $\Lambda_i$  with infidelity  $\varepsilon_i = 1 - F_e(\Lambda_i)$ , the product approximation expands as

$$1 - \widehat{F}^{\text{prod}}(\Lambda_{1:L}) = 1 - \prod_{i=1}^L (1 - \varepsilon_i) = \sum_{i=1}^L \varepsilon_i - \sum_{i < j} \varepsilon_i \varepsilon_j + \mathcal{O}(\varepsilon^3 L^3). \quad (7)$$

For Markovian compositions of Pauli-diagonal channels, the true infidelity shares the first-order term  $\sum_i \varepsilon_i$ . Differences between the true composed fidelity and the product approximation therefore enter through pairwise and higher-order terms. At the pair level this discrepancy scales as  $\sum_{i < j} \varepsilon_i \varepsilon_j$ , which is of order  $\varepsilon^2 L^2$  when the per-channel infidelities are comparable.

For non-Markovian compositions, the same scaling serves only as a warning diagnostic. If consecutive channels are coupled through a shared bath, each pair of time steps can contribute a cross-correlation term. A marginal estimator has no representation of these terms, so the number of possible pair corrections still grows like  $L^2$ . We therefore use  $\varepsilon^2 L^2$  only as an order-of-magnitude flag for when marginal factorisation should be stress-tested. Once  $\varepsilon$  is large or the bath coupling is non-perturbative, the expression is no longer a quantitative error estimate.

*Small  $\varepsilon$ : hardware-like noise.* In the device regime R1,  $\varepsilon \approx 10^{-3}$ . At  $L = 16$ , the diagnostic scale is  $\varepsilon^2 L^2 \approx 2.5 \times 10^{-4}$ , comparable to the residual error of the product approximation in our experiments. The correction that a learned model could recover is therefore smaller than the fitting error expected from a practical training set. This explains why the neural models do not improve over the analytic baseline. In this regime, the product approximation is the natural default, with direct measurement used when an unbiased experimental estimate is required.

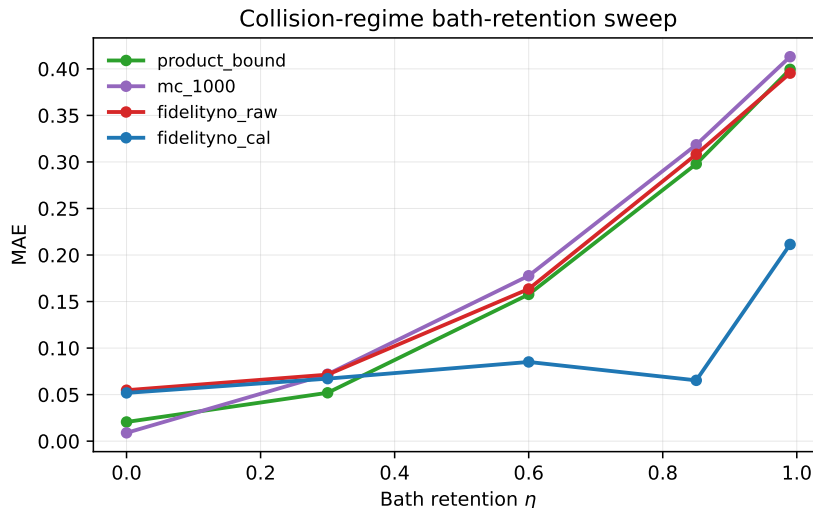
*Large  $\varepsilon$ : bath-mediated noise.* In the collision regime R2,  $\varepsilon \approx 0.4$ . At  $L = 16$ , the diagnostic scale is no longer perturbatively small; it exceeds the fidelity range and should be read as a warning that marginal factorisation has broken down. In this setting, bath-mediated correlations can dominate the error of marginal estimators. This is the regime where a learned surrogate can help, provided the deployment distribution is covered by training or by labelled-OOD recalibration. In our experiments, calibrated sequence models improve strongly over uncalibrated marginal predictors, but stratified DFE matches their MAE with about 32 shots per query. Their defensible advantage is therefore zero-measurement inference after an offline calibration cost has been amortised, not superior attainable accuracy.

Thus,  $\varepsilon^2 L^2$  is not a theorem-level error bound. It is a screening diagnostic: when it is small, there is little room for learning to improve on marginals; when it is large, marginal estimators should be stress-tested against correlated alternatives.

## 6.2. Bath-retention sweep

Figure 4 tests this picture by sweeping the bath-retention parameter  $\eta$  in the collision model. At  $\eta = 0$ , the bath is reset and the marginal description is nearly valid; MC-1000 and the product approximation have MAE 0.009 and 0.021, respectively. As  $\eta$  increases, the marginal estimators degrade quickly. At  $\eta = 0.85$ , product and MC-1000 reach MAE 0.298 and 0.318, whereas FIDELITYFORMER+cal remains at 0.065. At  $\eta = 0.99$ , FIDELITYFORMER+cal still improves over the marginal estimators, but its MAE rises to 0.211, indicating that this point lies near the edge of the calibrated training support.





**Figure 4. Bath-retention sweep for the decision rule.** Collision-regime MAE as the retention coefficient  $\eta$  interpolates between Markovian marginals ( $\eta = 0$ ) and strong memory ( $\eta \rightarrow 1$ ). Each point uses 1024 sequences. Neural curves are averaged over five seeds; analytic and MC-1000 curves are evaluated on the same split.

### 6.3. Decision rule

Table 6 summarises the resulting rule of thumb. The thresholds are calibrated to our three regimes and to the  $\eta$  sweep, and should be adjusted for other datasets.

**Table 6. Operating-regime decision rule.**

| Regime                                                         | Diagnostic signal                                     | Recommended estimator                                                     |
|----------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------|
| Weak correlations / low infidelity                             | $\varepsilon^2 L^2 \lesssim 10^{-3}$                  | Product approximation                                                     |
| Moderate correction, quantum shots affordable                  | $10^{-3} \lesssim \varepsilon^2 L^2 \lesssim 10^{-1}$ | DFE, with shot budget matched to accuracy target                          |
| Marginals valid and classical simulation affordable            | $\eta \approx 0$ or validated Markovian marginals     | MC with $K \in [10^2, 10^3]$                                              |
| Strong memory, stable shift, and many measurement-free queries | marginals fail; offline labels can be amortised       | calibrated low-capacity or neural surrogate                               |
| Strong memory without labels or shots                          | large raw OOD bias expected                           | raw surrogate only for ranking or triage, not absolute fidelity reporting |

The non-trivial cases are the middle and lower rows, where the deployment channel is correlated or non-Markovian. The choice is then a tradeoff among classical simulation cost, per-query quantum measurement cost, and one-time labelled characterisation cost. In our experiments, direct measurement is more accurate whenever 64 shots per query are acceptable. A learned surrogate becomes a cost option only when labelled OOD characterisation can be done offline, the shift remains stable, and enough later queries must be served without measurements to amortise that build cost.

#### 6.4. QEC compatibility

We do not claim that FIDELITYFORMER is a decoder, a threshold estimator, or a replacement for logical-error-rate estimation in QEC. It is better viewed as a proxy fidelity-estimation component in workflows where composed-channel quality must be queried repeatedly before committing to a more expensive decoder-level or experimental validation.

**Logical-channel quality estimation.** During code design, a logical-channel fidelity proxy can be useful for screening code patches or noise models. Our device regime suggests that, for low-infidelity Pauli-like noise, a learned surrogate is unlikely to improve on a calibrated analytic estimator. We do not infer logical error rates, decoder performance, or threshold behaviour from these fidelity estimates; those require separate decoder-level simulations or experiments.

**Memory cutoff and protocol selection.** At higher infidelity, as in the collision regime, the practical question is often whether to keep, abort, or re-purify a noisy memory or partially purified state. In such settings, FIDELITYFORMER+cal can provide repeated fidelity estimates without consuming experimental shots, provided the deployment regime is fixed and labelled calibration data can be obtained offline.

In both cases, the surrogate complements existing QEC tools rather than replacing them. Its use is justified only when the cost of standard simulation or measurement is larger than the cost of building and amortising an offline labelled set.

#### 6.5. Scope and caveats

Our claims are subject to four limitations.

- (i) *Labelled-OOD calibration requires labels.* The labelled calibration set ( $N_{\text{cal}} = 64$  in the headline configuration, with a sweep down to 8 in Fig. 2) must be obtained from exact simulation, an offline characterisation campaign, or another labelled-data source. The quantum or classical cost is paid once and amortised, but it is not free. We discuss unlabelled extensions, such as distribution matching at the predicted-quantile level, in Appendix Appendix B.
- (ii) *The collision memory strength is not identifiable from the input.* The reset-bath marginals are independent of  $\eta$  for fixed collision parameters. Predictions therefore

estimate a distribution-conditional correction; they do not reconstruct the realised bath state or per-sample retention coefficient.

- (iii) *The  $d = 4$  regime is not bath-mediated.* The two-qubit channels in R3 are order-sensitive, but their correlations are coherent rather than environmental. Whether the recalibration behaviour observed in R2 extends to genuinely bath-mediated systems with  $d > 2$  remains open.
- (iv) *The architecture is not the main claim.* FIDELITYFORMER is one member of a broader class of causal sequence models. Across our experiments, several learned backbones reach comparable accuracy within seed variation. The main contribution is the operating-regime characterisation and the calibration-based deployment picture, not a claim that this particular architecture is uniquely optimal.

## 7. Related work

*Fidelity-estimation tools.* The estimators compared in Section 4 are standard tools in quantum information. Product-style approximations appear in analyses of repeater chains and quantum-network protocols [1]. Direct Fidelity Estimation [5] estimates fidelity through Pauli measurements on the implemented channel. Randomised benchmarking [6, 17] estimates Clifford-averaged gate fidelity, while cross-entropy benchmarking [18] provides a related sequence-level diagnostic. Diamond-norm semidefinite programs [12, 19] give worst-case channel-distance estimates, at substantially higher classical cost.

*Learned surrogates for scientific computation.* Learned surrogates that replace an expensive numerical estimator with a fast forward model are now widespread across the sciences. Operator-learning architectures such as the Fourier Neural Operator [20] and DeepONet [21] approximate solution maps of parametric differential equations and have become standard tools in scientific machine learning. A recurring question in this literature is not which architecture is largest, but when a surrogate is trustworthy enough to replace the estimator it approximates, and how its error behaves out of distribution. Our study is an instance of this question for a composed-estimator prediction task: rather than proposing a new operator architecture, we characterise the operating regimes in which a learned surrogate is preferable to the analytic, sampling, or measurement-based estimator it is meant to replace, and quantify the labelled-calibration cost of closing the out-of-distribution gap.

*Learned channel-fidelity surrogates.* Neural surrogates have been used for quantum-dynamics prediction, dissipative evolution, and Lindbladian learning [22, 23]. Quantum-aware graph networks and equivariant variants have also been explored for quantum-network estimation [24, 25]. In the repeater setting, related work includes reinforcement-learning agents for protocol selection [26] and Bayesian-optimisation approaches to

protocol search [27]. Our focus is different: we ask when a learned surrogate is preferable to analytic, Monte-Carlo, or measurement-based estimators, and when it is not.

*Non-Markovian collision models.* Collision models are widely used to study non-Markovian open-system dynamics [8, 28]. Prior work mainly uses them as forward-simulation models. Here we use a collision-model family as a stress test for fidelity estimators that only see marginal channel information.

*Distribution shift and post-hoc calibration.* Post-hoc calibration methods such as affine, temperature, and isotonic calibration are standard in classification and probabilistic regression [29, 30]. Our labelled-OOD calibration setting is closer to bias correction than to classical coverage calibration: the goal is to remove a systematic shift in the median fidelity prediction. Conformal prediction [31] could be used to calibrate predictive intervals for  $F_e$  and is a natural extension.

*Surrogate-driven QEC code design.* Learned surrogates have recently been proposed inside QEC code-design loops as inner-loop fidelity proxies [32]. Our results suggest a regime-dependent interpretation of such methods, but not a decoder-level or threshold-level claim. In low-infidelity Pauli-like regimes, analytic estimators may already be near-optimal; learned surrogates become more relevant when correlated noise produces corrections that marginal estimators cannot represent.

## 8. Conclusion

We asked when a learned surrogate is justified for composed-channel fidelity prediction. The strongest answer is negative: with full Choi descriptions of the studied Markovian  $d \leq 4$  sequences, exact superoperator composition is both numerically exact and already competitive in wall-clock time. In the non-Markovian collision stress test, observable marginals omit the bath memory, but an optimised measurement baseline is also much stronger than a with-replacement implementation suggests: stratified DFE matches the calibrated neural error with about 32 shots and surpasses it with 64.

The remaining niche is repeated measurement-constrained deployment under a stable distribution shift. Same-budget controls show a sequence-model advantage at 64 labelled OOD examples, while a low-capacity summary ridge catches up by 256 labels. Finite-shot calibration shows that 4096 offline shots can preserve the neural MAE and are amortised after about 128 queries relative to 32-shot DFE. Learned fidelity estimation is therefore an amortisation strategy, not a universal replacement for exact composition, classical simulation, or direct measurement.

*Future work.* Three directions follow from this analysis. First, labelled-OOD calibration could be replaced by unlabelled distribution matching, for example at the predicted-quantile level. Second, the same regime pattern should be tested in

genuinely bath-mediated systems beyond  $d = 2$ , such as three-qubit collision models. Third, tomography-aware encoders may improve the surrogate itself; for example, a Stinespring-isometry-aware encoder could represent environment-purification structure more directly than the Choi encoder used here.

## Acknowledgments

This study was funded by the Innovation Program for Quantum Science and Technology (Grant No. 2021ZD0302900). The funder played no role in study design, data collection, analysis and interpretation of data, or the writing of this manuscript.

## Author contributions

SW and XL contributed equally to this work. SW and XL conceived the study, developed the methodology, performed the experiments, analysed the results, and prepared the figures and tables. SW drafted the manuscript. WY supervised the project, guided the interpretation and framing of the results, and revised the manuscript. All authors discussed the results, commented on the manuscript, and approved the final version.

## Competing interests

All authors declare no financial or non-financial competing interests.

## Data availability

All datasets are generated synthetically by the scripts in the public repository, <https://github.com/shchng9963-droid/fidelityno>. Versioned generated-data and result archives will be deposited with an archival DOI before acceptance; the submission bundle records the exact code revision and random seeds.

## Code availability

The underlying code for this study is available in Github and can be accessed via this link <https://github.com/shchng9963-droid/fidelityno>.

## Use of AI and AI-assisted technologies

LLMs were used to improve grammar, wording, and readability of the manuscript text.

- [1] Azuma K, Economou S E, Elkouss D, Hilaire P, Jiang L, Lo H K and Tzitrin I 2023 *Reviews of Modern Physics* **95** 045006
- [2] Dür W and Briegel H J 2007 *Reports on Progress in Physics* **70** 1381–1424
- [3] Cross A W, Bishop L S, Sheldon S, Nation P D and Gambetta J M 2019 *Physical Review A* **100** 032328
- [4] Roffe J 2019 *Contemporary Physics* **60** 226–245

- [5] Flammia S T and Liu Y K 2011 *Physical review letters* **106** 230501
- [6] Helsen J, Xue X, Vandersypen L M and Wehner S 2019 *npj Quantum Information* **5** 71
- [7] Javadi-Abhari A, Treinish M, Krsulich K, Wood C J, Lishman J, Gacon J, Martiel S, Nation P D, Bishop L S, Cross A W *et al.* 2024 *arXiv preprint arXiv:2405.08810*
- [8] Ciccarello F, Lorenzo S, Giovannetti V and Palma G M 2022 *Physics Reports* **954** 1–70
- [9] Choi M D 1975 *Linear algebra and its applications* **10** 285–290
- [10] Jamiolkowski A 1972 *Reports on mathematical physics* **3** 275–278
- [11] Schumacher B 1996 *Physical Review A* **54** 2614
- [12] Watrous J 2013 *Chicago Journal OF Theoretical Computer Science* **8** 1–19
- [13] Diamond S and Boyd S 2016 *Journal of Machine Learning Research* **17** 1–5
- [14] Lambert N, Giguère E, Menczel P, Li B, Hopf P, Suárez G, Gali M, Lishman J, Gadhvi R, Agarwal R *et al.* 2026 *Physics Reports* **1153** 1–62
- [15] Wood C J, Biamonte J D and Cory D G 2015 *Quantum Information & Computation* **15** 759–811
- [16] Zaheer M, Kottur S, Ravanbakhsh S, Poczos B, Salakhutdinov R R and Smola A J 2017 *Advances in neural information processing systems* **30**
- [17] Helsen J, Roth I, Onorati E, Werner A H and Eisert J 2022 *PRX quantum* **3** 020357
- [18] Boixo S, Isakov S V, Smelyanskiy V N, Babbush R, Ding N, Jiang Z, Bremner M J, Martinis J M and Neven H 2018 *Nature Physics* **14** 595–600
- [19] Khatri S, Sharma K and Wilde M M 2020 *Physical Review A* **102** 012401
- [20] Li Z, Kovachki N, Azizzadenesheli K, Liu B, Bhattacharya K, Stuart A and Anandkumar A 2021 Fourier neural operator for parametric partial differential equations *International Conference on Learning Representations (ICLR)*
- [21] Lu L, Jin P, Pang G, Zhang Z and Karniadakis G E 2021 *Nature Machine Intelligence* **3** 218–229
- [22] Herrera Rodriguez L E and Kananenka A A 2021 *The journal of physical chemistry letters* **12** 2476–2483
- [23] Ivashkov P, Romanov N, Gong W, Gu A, Hu H Y and Yelin S F 2026 *arXiv preprint arXiv:2603.05492*
- [24] Verdon G, McCourt T, Luzhnica E, Singh V, Leichenauer S and Hidary J 2019 *arXiv preprint arXiv:1909.12264*
- [25] Forestano R T, Comajoan Cara M, Dahale G R, Dong Z, Gleyzer S, Justice D, Kong K, Magorsch T, Matchev K T, Matcheva K *et al.* 2024 *Axioms* **13** 160
- [26] Reiß S D and van Loock P 2023 *Physical Review A* **108** 012406
- [27] La Corte L, Goodenough K, Maity A G, Santra S and Elkouss D 2025 Bayesian optimization for repeater protocols 2025 *International Conference on Quantum Communications, Networking, and Computing (Q CNC)* (IEEE) pp 135–142
- [28] Filippov S N and Luchnikov I A 2022 *Physical Review A* **105** 062410
- [29] Guo C, Pleiss G, Sun Y and Weinberger K Q 2017 On calibration of modern neural networks *International conference on machine learning* (PMLR) pp 1321–1330
- [30] Kuleshov V, Fenner N and Ermon S 2018 Accurate uncertainties for deep learning using calibrated regression *International conference on machine learning* (PMLR) pp 2796–2804
- [31] Shafer G and Vovk V 2008 *Journal of machine learning research* **9**
- [32] Varbanov B M, Serra-Peralta M, Byfield D and Terhal B M 2025 *Physical Review Research* **7** 013029

## Appendix A. First-order accuracy of the product approximation for Pauli-diagonal channels

We sketch why the product approximation (1) is first-order accurate when each per-channel marginal  $\Lambda_i$  is Pauli-diagonal in a common basis, which is the configuration realised by our IBM **Fake\*V2** noise-model regime to leading order. The result we use is a scaling argument  $\hat{F}_{\text{err}}^{\text{prod}} = \mathcal{O}(\varepsilon^2 L^2)$ , not a saturation theorem.

Let  $\Lambda_i$  act as a Pauli channel,

$$\Lambda_i(\rho) = \sum_{P \in \{I, X, Y, Z\}^{\otimes \log_2 d}} p_{i,P} P \rho P^\dagger, \quad (\text{A.1})$$

with  $\sum_P p_{i,P} = 1$ . The Choi matrix  $\mathcal{J}(\Lambda_i)$  is then diagonal in the Pauli basis, and the entanglement fidelity reduces to the identity weight  $F_e(\Lambda_i) = p_{i,I}$ .

Composition  $\Lambda_{1:n} = \Lambda_n \circ \dots \circ \Lambda_1$  produces another Pauli channel. The identity weight of the composed channel is *not* in general equal to the product  $\prod_i p_{i,I}$  of identity weights: contributions of the form  $PPPP \dots = I$  for the same Pauli  $P$  at multiple positions, and triple-products  $P_1 P_2 P_3 = I$ , also land in the identity bucket. Counting cross-channel pair contributions of  $(P, P)$  for  $P \neq I$  at any pair of positions gives a correction

$$p_{1:n,I} - \prod_{i=1}^n p_{i,I} = \sum_{i < j} \sum_{P \neq I} p_{i,P} p_{j,P} \prod_{k \neq i,j} p_{k,I} + \mathcal{O}(L^3 \varepsilon^3). \quad (\text{A.2})$$

Each summand is  $\mathcal{O}(\varepsilon^2)$  and there are  $\binom{n}{2}$  pairs, giving a leading-order correction of order  $\varepsilon^2 L^2$  where  $L = n$  and  $\varepsilon$  is the typical per-channel infidelity. The product approximation is therefore first-order accurate in  $\varepsilon$  but acquires a  $\varepsilon^2 L^2$  correction at second order.

When the channels are not Pauli-diagonal in a common basis, an additional ‘‘Pauli-rotation’’ correction arises but it remains  $\mathcal{O}(\varepsilon^2)$  at leading order. The collision-model regime exhibits a much larger correction because the dynamics are not Pauli at all, and the bath-mediated terms are  $\mathcal{O}(\eta)$  rather than  $\mathcal{O}(\varepsilon^2)$ ; this is what the surrogate learns to recover.

We do not require this scaling for any of the experimental claims in the main text; it is included here to clarify the precise scope of the ‘‘product approximation is near-exact’’ statement in (R1).

## Appendix B. Affine recalibration across random seeds

Table B1 reports a separate legacy audit in which per-seed affine  $(a, b)$  coefficients were fitted on a labelled, held-out 10% subset of the collision `family_ood` (memory-strength-`OOD`) data. Its calibration set and split realisation differ from the fixed  $N_{\text{cal}} = 64$  comparison in Table 2, so its mean MAE of 0.094 should not be used as a paired comparison with the headline 0.091. Within this audit, the five seeds reach similar calibrated MAE (std  $1.3 \times 10^{-3}$ ) despite variation in the fitted affine coefficients (slope std  $\approx 0.09$ ), indicating that the slope and intercept are correlated and should not be interpreted separately.

**Table B1. Affine recalibration coefficients, FidelityFormer on collision family\_ood.** 5 seeds, 10% OOD calibration set. The fitted slope is  $a$  and intercept is  $b$  in  $\hat{F}^{\text{cal}} = a\hat{F}^{\text{raw}} + b$ .

| seed | $a$   | $b$    | MAE raw | MAE cal      |
|------|-------|--------|---------|--------------|
| 0    | 2.845 | −1.916 | 0.387   | 0.095        |
| 1    | 2.822 | −1.896 | 0.386   | 0.095        |
| 2    | 2.918 | −1.980 | 0.387   | 0.094        |
| 3    | 3.026 | −2.080 | 0.388   | 0.095        |
| 4    | 3.003 | −2.052 | 0.387   | 0.092        |
| mean | 2.923 | −1.985 | 0.387   | 0.094        |
| std  | 0.091 | 0.081  | 0.001   | <b>0.001</b> |

*Seed-to-seed variation in affine coefficients.* The  $\sim 0.1$  variation in the slope  $a$  does not translate into comparable variation in test MAE because slope and intercept co-vary over the narrow range of raw predictions. The result supports reporting predictive error across seeds, but it does not by itself identify the physical source of the OOD bias.

*Quantile calibration.* We apply the same affine  $(a, b)$  to all  $K = 9$  predicted quantile levels, which preserves quantile monotonicity but not necessarily empirical coverage on OOD. In our experiments quantile coverage is recovered to within  $\pm 0.04$  of nominal at the medium-to-high quantiles ( $q \geq 0.5$ ); coverage at low quantiles ( $q \leq 0.3$ ) is occasionally over-coverage on family\_ood due to clipping at  $F_e = 0$ . A more principled extension would fit a per-quantile shift, which we leave to future work.

## Appendix C. Implementation details

*Software and numerical backends.* PyTorch 2.6 + CUDA 12.8 (RTX 5090, sm\_120). Differentiable simulator: `dynamiqs` 0.3 for time-dependent Lindblad evolution, with `qutip` 5.2 used for cross-validation of Choi matrices to a tolerance of  $10^{-8}$ . Diamond-norm SDP via `cvxpy` 1.9.1 wrapping the SCS solver, called through `qutip.dnorm` for production runs. Hydra for configs; `wandb` for offline logging.

*Model architecture.* FIDELITYFORMER:  $d_{\text{model}} = 256$ , depth 6, heads 8, FFN expansion  $4\times$ , rotary positional encoding, GELU non-linearity, dropout 0.1. Channel encoder: 2-layer MLP, hidden dim 512, GELU. Quantile head:  $K = 9$  levels at  $\{0.1, 0.2, \dots, 0.9\}$ , softplus output, bias initialised so that the median quantile starts at  $F_e \approx 0.7$  (essential to avoid dead-gradient saturation in the deep-circuit regime). Auxiliary physics-consistency head: 2-layer MLP, 0.1 weight on auxiliary loss.



*Training protocol.* AdamW, lr  $3 \times 10^{-4}$ , cosine decay to  $10^{-5}$ , weight decay  $10^{-2}$ . 80 epochs, batch size 256 (collision and  $d=4$ ), batch size 128 (device regime). Early-stopping patience 20. We apply a length curriculum: starting epochs sample lengths  $\in \{2, 4\}$  uniformly, expanding to  $\{2, 4, 8, 16\}$  over the first 30 epochs.

Architecture and training hyperparameters were selected on the calibration split of the device regime and held fixed across regimes. We did not run a per-regime sweep, which we view as a stress test of the architecture’s regime-agnosticity.

*Dataset splits.* Each split contains 8,000 sequences (collision,  $d=4$ ) or 1,024 sequences per backend (device). Lengths are sampled from  $\{2, 4, 8, 16\}$  for train/calib/id\_test and  $\{24, 32, 48\}$  for length\_ood. In the collision data, the legacy family\_ood filename denotes a held-out  $\eta \in [0.85, 0.99]$  interval; training uses  $\eta \in [0, 0.7]$ . Random seeds are disjoint across splits.

*Reproducibility and validation.* The five seeds are  $\{0, 1, 2, 3, 4\}$ , applied to (a) NumPy / PyTorch global RNG, (b) channel parameter sampling, (c) train/val/test split assignment, (d) model initialisation, (e) data shuffling. The 84 unit tests cover CPTP sanity for every channel-family generator, Choi composition associativity, quantile-head monotonicity, recalibration coefficient stability, diamond-SDP correctness against a QuTiP reference, and an end-to-end smoke-train on a 100-sample subset. Total wall-clock for a fresh reproduction on a  $2 \times$  RTX 5090 setup:  $\sim 5$  hours, dominated by neural training.